

Harvesting fish



FISH ARE THE MOST popular kind of seafood, with some 77 million tons (70 million metric tons) caught around the world each year. Some fish are caught by hand-thrown nets and traps in local waters, but far more are caught at sea by modern fishing vessels using the latest technology. Some fish are caught on long

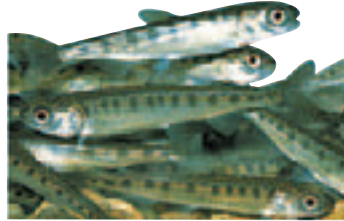
lines with many hooks or ensnared when they swim into long walls of drift nets. Bottom-dwelling fish are trawled or whole shoals are gathered up in huge nets set in midwater. Using sonar to detect shoals means there are few places where fish can escape notice. Even fish living in deep waters, such as orange roughy at depths of 3,300 ft (1,000 m), are brought up in numbers. Many people are concerned that too many fish are being caught because numbers take a long time to recover. Competition for fish stocks is fierce and it is difficult for fishermen to make a living. But some fish, such as salmon, are farmed to help meet demand.

3 AT SEA

Atlantic salmon spend up to four years at sea, feeding on other fishes. They grow rapidly, putting on several pounds annually. Then the mature salmon return to their home rivers and streams where they hatch. They recognize their home stream by a number of clues, including its "smell" (particular combinations of tiny quantities of substances in the water).

1 HATCHING OUT

Salmon begin life in rivers and streams where they hatch from eggs laid in a shallow hollow among gravel. First the fry (alevins) grow, using the contents of their egg sac attached to their bellies as food.



2 YOUNG SALMON At a few weeks old, the egg sac disappears, so young salmon must feed on tiny insects in the river. Soon dark spots appear on the young salmon (called parr). The parr stay in the river for a year or more, before turning into silvery smolt which head for the sea.

Fin rays are well-developed

Large, first dorsal fin

Pelvic fin

Pectoral fin

Operculum (flap covering gills)

Mouth for feeding and taking in water to "breathe"

FISH FARMING

Salmon are among the few kinds of sea fish to be farmed successfully. Young salmon are reared in freshwater then, when large enough, released into floating pens in calm sea waters, such as sea lochs. To help them grow quickly, the salmon are fed with dried fish pellets. Environmentalists are concerned that fish parasites, called sea lice, common among farmed salmon, are infecting and killing wild salmon. Biologists are investigating ways to combat this problem.

